

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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1 Introduction

The release of ChatGPT in November 2022 marked the beginning of large-scale diffusion of large language models (LLMs) around the world. Within months, millions of workers gained access to tools capable of drafting documents, writing code, analyzing data, and performing countless other cognitive tasks that previously required human expertise. This rapid integration into workplace workflows has generated widespread anxiety about the future of work, technological unemployment, and economic disruption, with some early evidence showing large productivity gains (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023).

This uncertainty reflects a potential shift in the nature of this technological innovation. Unlike previous technologies that predominantly displaced routine tasks in lower- and middle-skilled occupations (Autor, 2015; Autor et al., 2003), LLMs possess capabilities that extend across the entire skill distribution, with the potential to affect highly skilled professional and analytical roles as well (Eloundou et al., 2024). Estimates suggest that around 80% of the U.S. workforce could see at least 10% of their tasks affected, with exposure rising in higher-income occupations on average (Eloundou et al., 2024). As such, the introduction of generative AI raises important questions about whether LLMs will erode job security, depress white-collar wages, and increase or decrease inequality.

To examine these questions, we use detailed UK labor market data from Revelio Labs, covering millions of firms and job postings between 2021 and 2025. Our identification strategy exploits the sudden and arguably exogenous introduction of ChatGPT in November 2022. The impact of this shock varied across firms and industries depending on their occupational mix and the extent to which core activities are susceptible to LLM automation. We measure this heterogeneity using the occupational level LLM exposure scores developed by Eloundou et al. (2024), who evaluate 19,265 detailed work activities from the O*NET database using both human annotators and GPT-4 to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. These exposure scores allow us to quantify the share of tasks within each occupation, or the share of jobs within each firm that can be performed by LLMs.¹

¹Their measure strongly correlates with other LLM exposure measures (see e.g. Tomlinson et al., 2025; Hampole et al., 2025).